WORKED SOLUTION

# STEP III 1995, Question 14

Probability and Statistics • Moment Generating Functions

**STEP III 1995, Question 14**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q14

## The key idea

The moment generating function turns sums of independent random variables into products: if $T_1$ and $T_2$ are independent then $\mathrm{E}[\mathrm{e}^{\theta(T_1+T_2)}] = \mathrm{E}[\mathrm{e}^{\theta T_1}]\,\mathrm{E}[\mathrm{e}^{\theta T_2}]$. So we find one MGF, raise it to a power, and then recognise the result as the MGF of a density of the same family. Because an MGF determines its distribution, recognising it is enough.

## Solution

### The MGF of $T$

For a real number $\theta$,

$$M_T(\theta) = \mathrm{E}[\mathrm{e}^{\theta T}] = \int_0^\infty t\,\mathrm{e}^{-(1-\theta)t}\,\mathrm{d}t.$$

This converges only when $\theta < 1$ (otherwise the integrand does not tend to 0), so we assume $\theta < 1$ from now on. The integral we need repeatedly is

$$\int_0^\infty t^k \mathrm{e}^{-\lambda t}\,\mathrm{d}t = \frac{k!}{\lambda^{k+1}} \qquad (\lambda > 0,\ k = 0, 1, 2, \dots),$$

which follows by integrating by parts $k$ times (or by substituting $s = \lambda t$ and using $\int_0^\infty s^k \mathrm{e}^{-s}\,\mathrm{d}s = k!$). With $k = 1$ and $\lambda = 1 - \theta > 0$,

$$\boxed{M_T(\theta) = \frac{1}{(1-\theta)^2}}, \qquad \theta < 1.$$

As a quick sanity check, $M_T(0) = 1$, as every MGF must satisfy.

### The MGF of $U$

Let $T_1$ and $T_2$ be the times for the two questions, so $U = T_1 + T_2$, where $T_1$ and $T_2$ each have density f and are independent (as the question tells us). Then $\mathrm{e}^{\theta T_1}$ and $\mathrm{e}^{\theta T_2}$ are

independent too, so the expectation of their product is the product of their expectations:

$$M_U(\theta) = \mathrm{E}[\mathrm{e}^{\theta T_1}\mathrm{e}^{\theta T_2}] = M_T(\theta)^2 = \frac{1}{(1-\theta)^4}, \qquad \theta < 1.$$

#### The density of $U$

Now we compute the MGF of the proposed density $\mathrm{g}(u) = \frac{1}{6}u^3\mathrm{e}^{-u}$. Using the integral above with $k = 3$ and $\lambda = 1 - \theta$,

$$\int_0^\infty \mathrm{e}^{\theta u}\,\tfrac{1}{6}u^3\mathrm{e}^{-u}\,\mathrm{d}u = \frac{1}{6}\cdot\frac{3!}{(1-\theta)^4} = \frac{1}{(1-\theta)^4}.$$

(Setting $\theta = 0$ confirms that g integrates to 1, so it is a genuine density.)

**Ask yourself** We have shown that g has the same MGF as $U$. Why does that mean $U$ has density g?

This uses the uniqueness theorem for MGFs: if two random variables have MGFs that exist and agree on an open interval around $\theta = 0$, they have the same distribution. Here the two agree for every $\theta < 1$, which includes the interval $(-1, 1)$. So $U$ has density

$$\mathrm{g}(u) = \frac{1}{6}u^3\mathrm{e}^{-u}, \qquad u \geq 0,$$

as required.

If you would rather not use the uniqueness theorem, the convolution formula gives the same result directly. Since $T_1, T_2 \geq 0$, for $u \geq 0$,

$$\begin{aligned}\mathrm{g}(u) &= \int_0^u \mathrm{f}(s)\,\mathrm{f}(u-s)\,\mathrm{d}s = \mathrm{e}^{-u}\int_0^u s(u-s)\,\mathrm{d}s \\ &= \mathrm{e}^{-u}\left[\tfrac{1}{2}us^2 - \tfrac{1}{3}s^3\right]_0^u = \frac{1}{6}u^3\mathrm{e}^{-u}.\end{aligned}$$

#### The total time for $n$ questions

Let $U_n = T_1 + \cdots + T_n$, where the $T_i$ are independent with density f. The same product argument gives

$$M_{U_n}(\theta) = M_T(\theta)^n = \frac{1}{(1-\theta)^{2n}}, \qquad \theta < 1.$$

The pattern from $n = 2$ suggests the density $\mathrm{h}_n(u) = c\,u^{2n-1}\mathrm{e}^{-u}$. Its MGF is $c\,(2n-1)!/(1-\theta)^{2n}$, by the integral with $k = 2n-1$. This equals $M_{U_n}$ exactly when $c = 1/(2n-1)!$. By uniqueness again, the total time for $n$ questions has density

$$\boxed{\mathrm{h}_n(u) = \frac{u^{2n-1}\mathrm{e}^{-u}}{(2n-1)!}, \qquad u \geq 0.}$$

For $n = 1$ this gives f, and for $n = 2$ it gives g, as it should.

## Checks

- $\mathrm{E}[T] = M_T'(0) = 2$, which matches $\int_0^\infty t^2 \mathrm{e}^{-t}\,\mathrm{d}t = 2$. So $\mathrm{E}[U_n] = 2n$. The same value comes from $\mathrm{h}_n$: $\int_0^\infty u^{2n}\mathrm{e}^{-u}\,\mathrm{d}u/(2n-1)! = (2n)!/(2n-1)! = 2n$.
- Numerically, Simpson's rule gives $\int_0^\infty \mathrm{e}^{0.5t}\,t\mathrm{e}^{-t}\,\mathrm{d}t = 4.000$, matching $1/(1-0.5)^2$. A simulation of $2 \times 10^5$ sums of three times drawn from f gave $\mathrm{P}(U_3 \leq 5) \approx 0.386$, against $\int_0^5 u^5\mathrm{e}^{-u}/120\,\mathrm{d}u \approx 0.384$ from $\mathrm{h}_3$.

## Common slips

- Forgetting the restriction $\theta < 1$: the MGF does not exist for $\theta \geq 1$.
- Adding MGFs instead of multiplying them. The MGF of $U$ is $M_T(\theta)^2$, not $2M_T(\theta)$. (The MGF of $2T$ would be $M_T(2\theta)$, which is different again.)
- Getting the factorial in the normalising constant wrong. The power of $u$ is $2n-1$, so the constant is $1/(2n-1)!$, not $1/(2n)!$.

## Beyond STEP

The density $t\mathrm{e}^{-t}$ is the gamma distribution with shape 2 and rate 1, and $\mathrm{h}_n$ is the gamma distribution with shape $2n$. This question shows that independent gammas with the same rate add by adding their shapes. In particular, each $T$ behaves like the sum of two independent exponential times, which is how the gamma distribution arises as the waiting time for the $k$th event in a Poisson process. At university you will meet the characteristic function $\mathrm{E}[\mathrm{e}^{\mathrm{i}\theta X}]$, which always exists, and use it to prove the central limit theorem. Grimmett and Welsh, *Probability: An Introduction*, develops generating functions and these ideas clearly.

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